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(54) **SYSTEMS AND METHODS FOR
AUTOMATED PROGRAMMING OF
ELECTRICAL STIMULATION**

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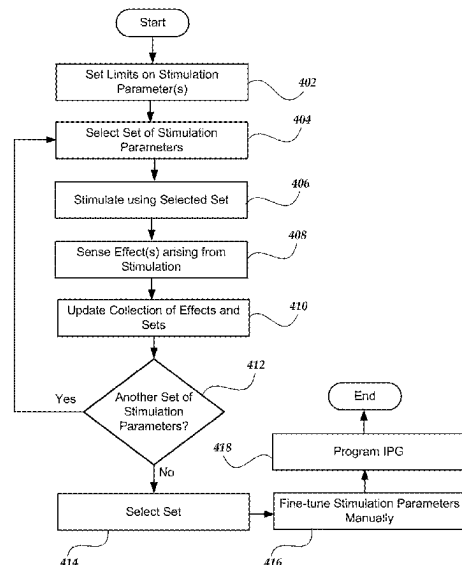
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(57) **ABSTRACT**

A method for automating selection of stimulation parameters
for a stimulation device implanted in a patient includes
setting, by a user, at least one limit on each of at least one
stimulation parameter and performing, automatically using
at least one processor, the following actions for each of a
plurality of sets of the stimulation parameters constrained by
the at least one limit: stimulating the patient, by the stimu-
lation device, using the set of stimulation parameters, sens-
ing one or more effects arising in response to the stimulation,
and updating, by the at least one processor, a collection of
the effects and sets of stimulation parameters with the one or
more effects and the set of stimulation parameters. The
method further includes selecting, by the processor, one of
the sets of stimulation parameters based on the effects.

18 Claims, 7 Drawing Sheets



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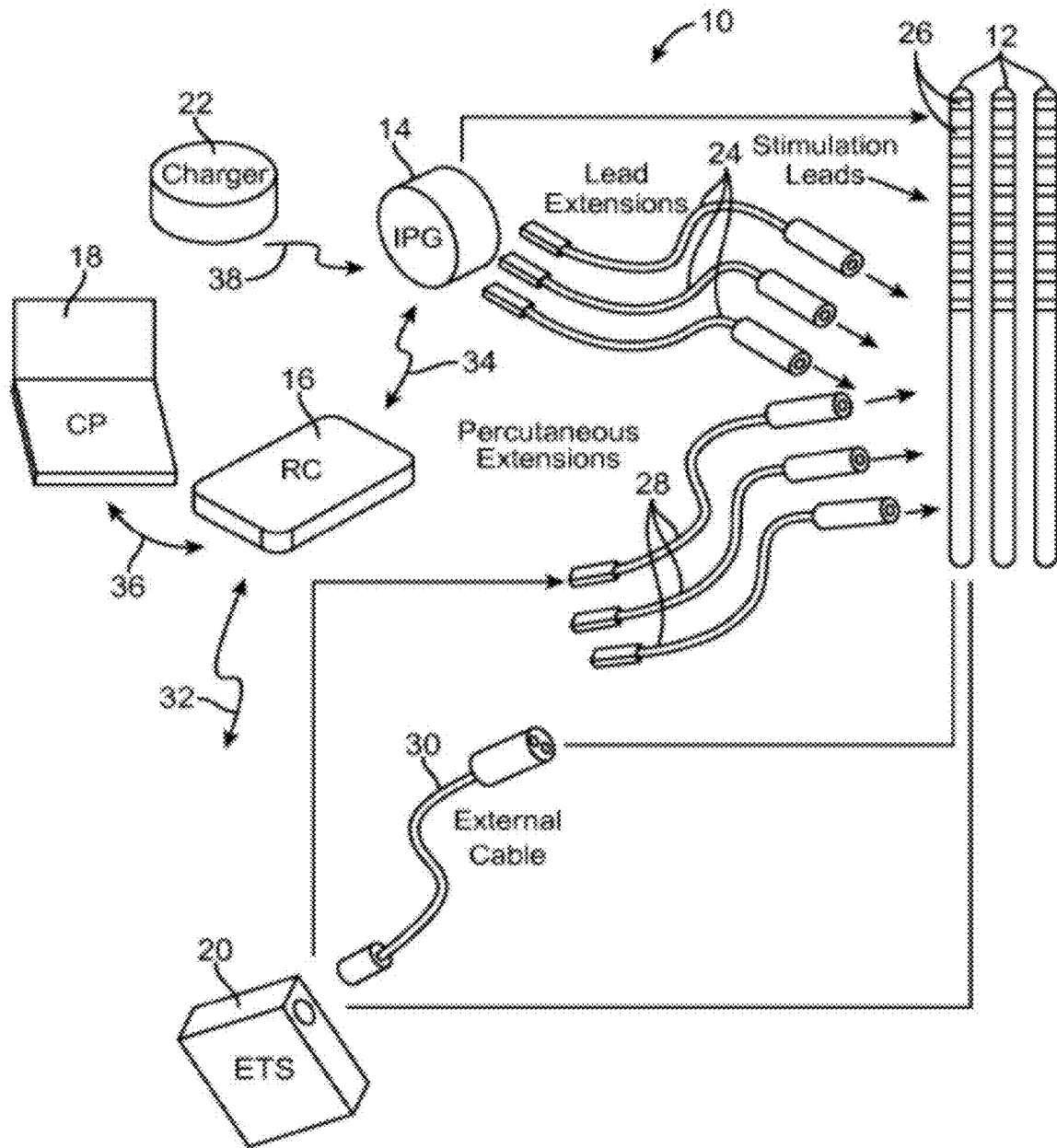
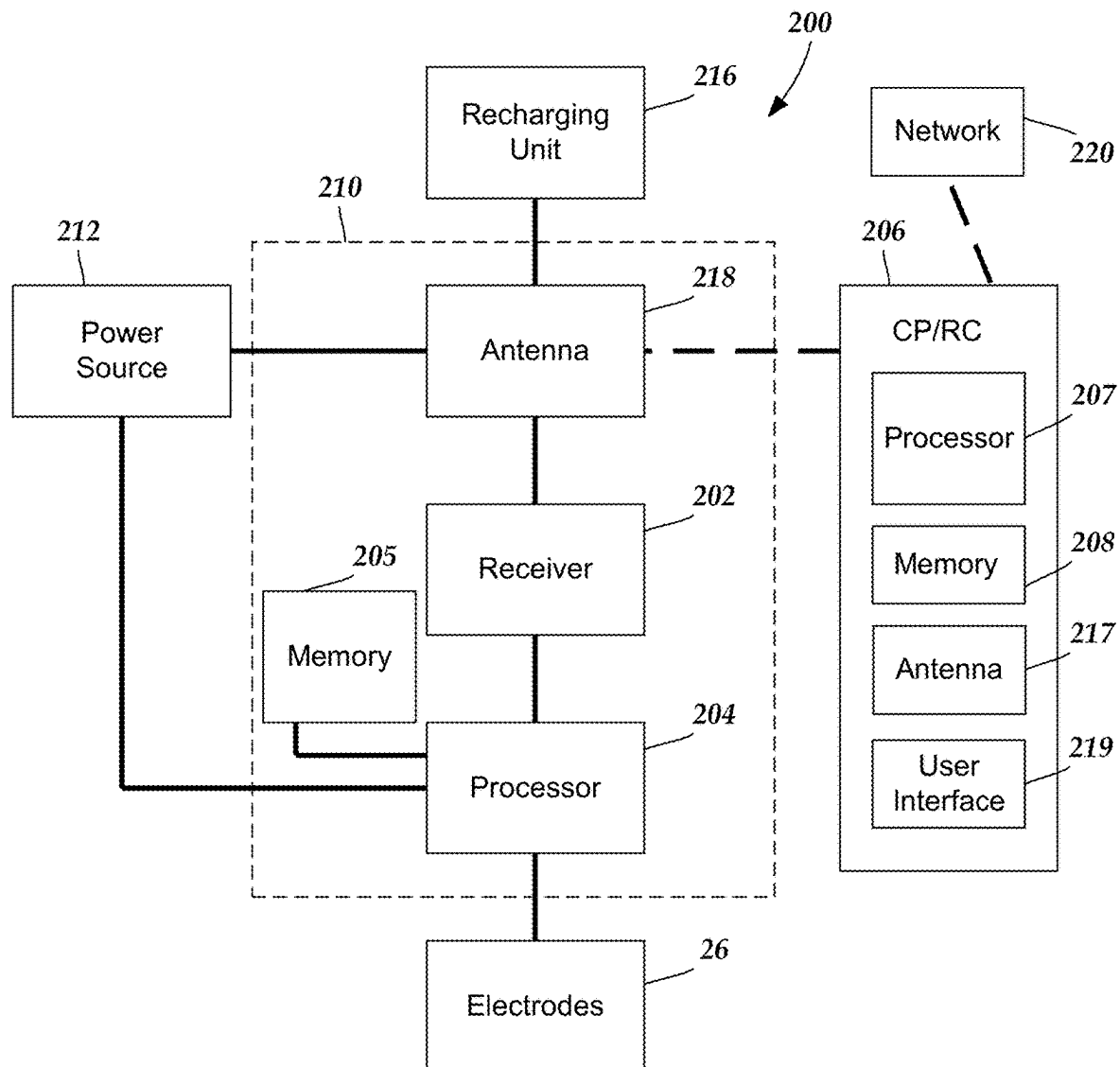
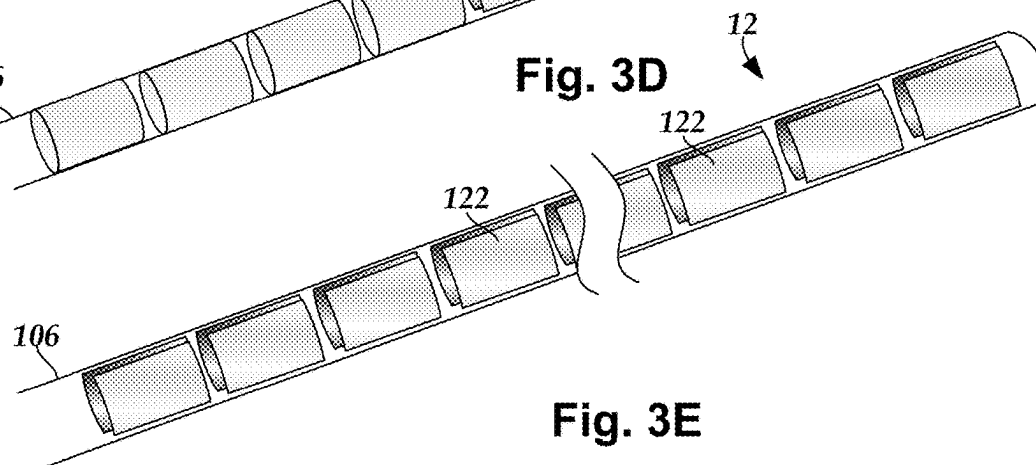
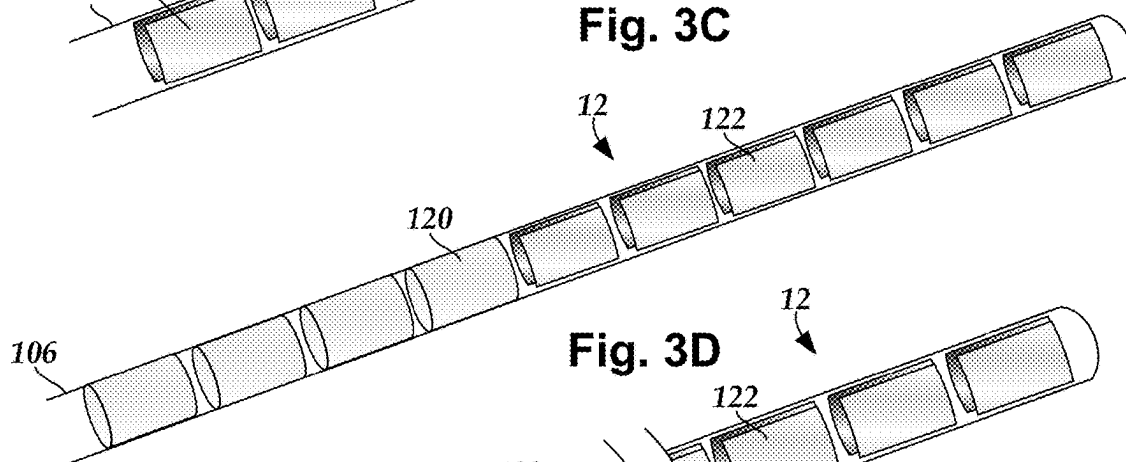
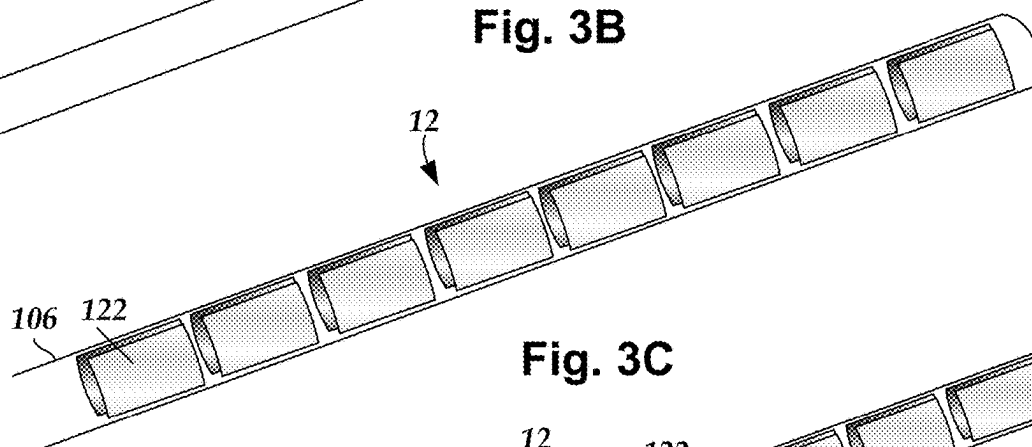
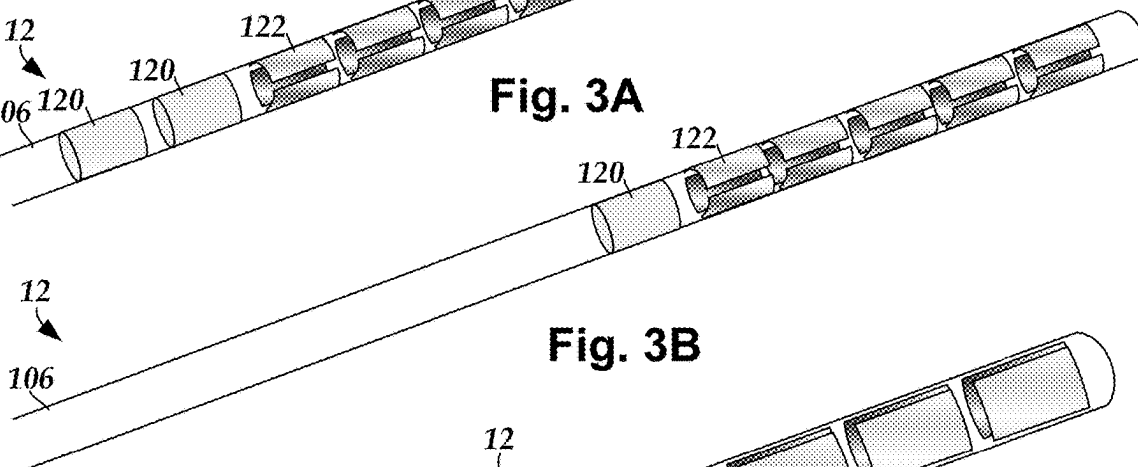
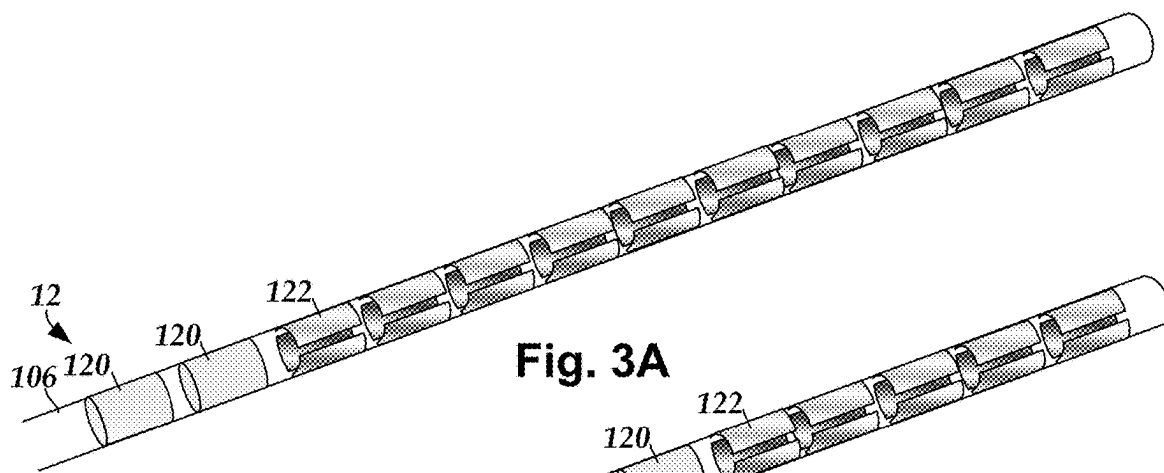
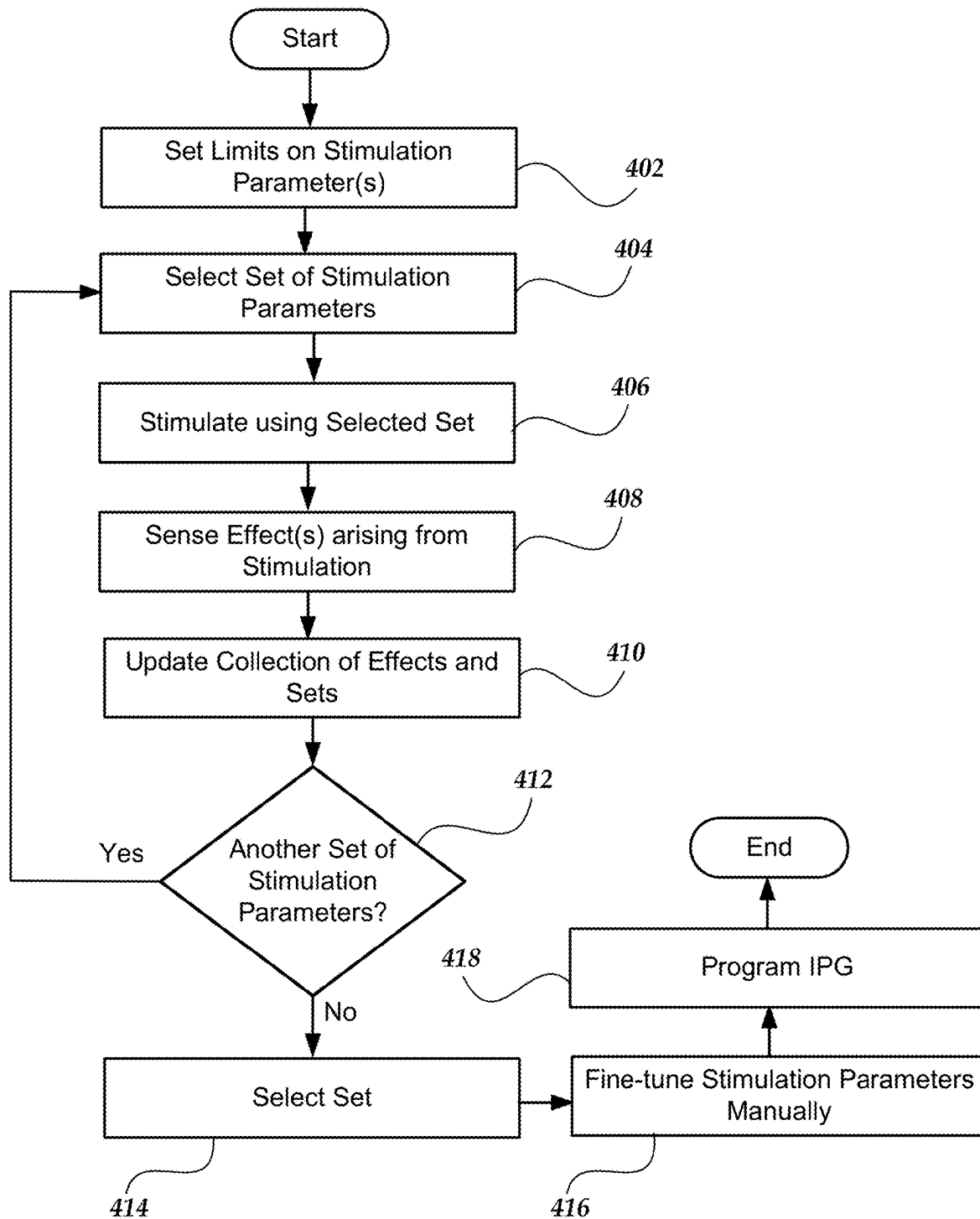
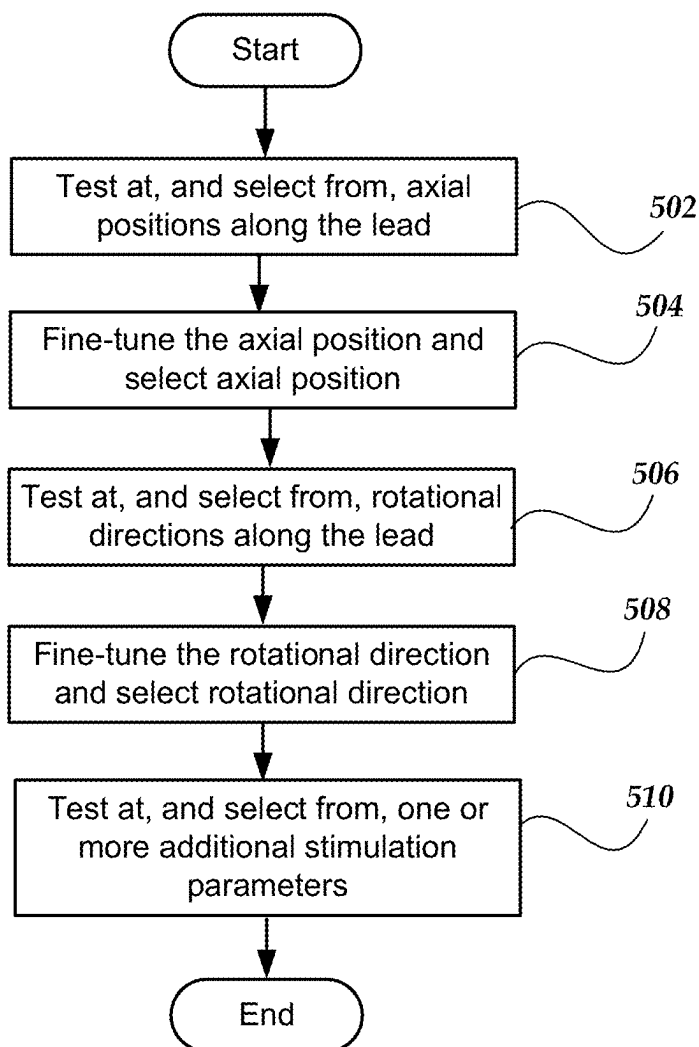
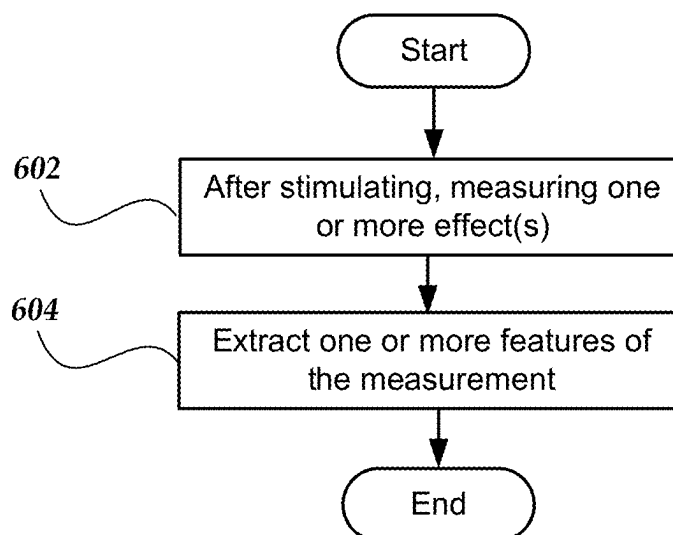


Fig. 1

**Fig. 2**



**Fig. 4**

**Fig. 5****Fig. 6**

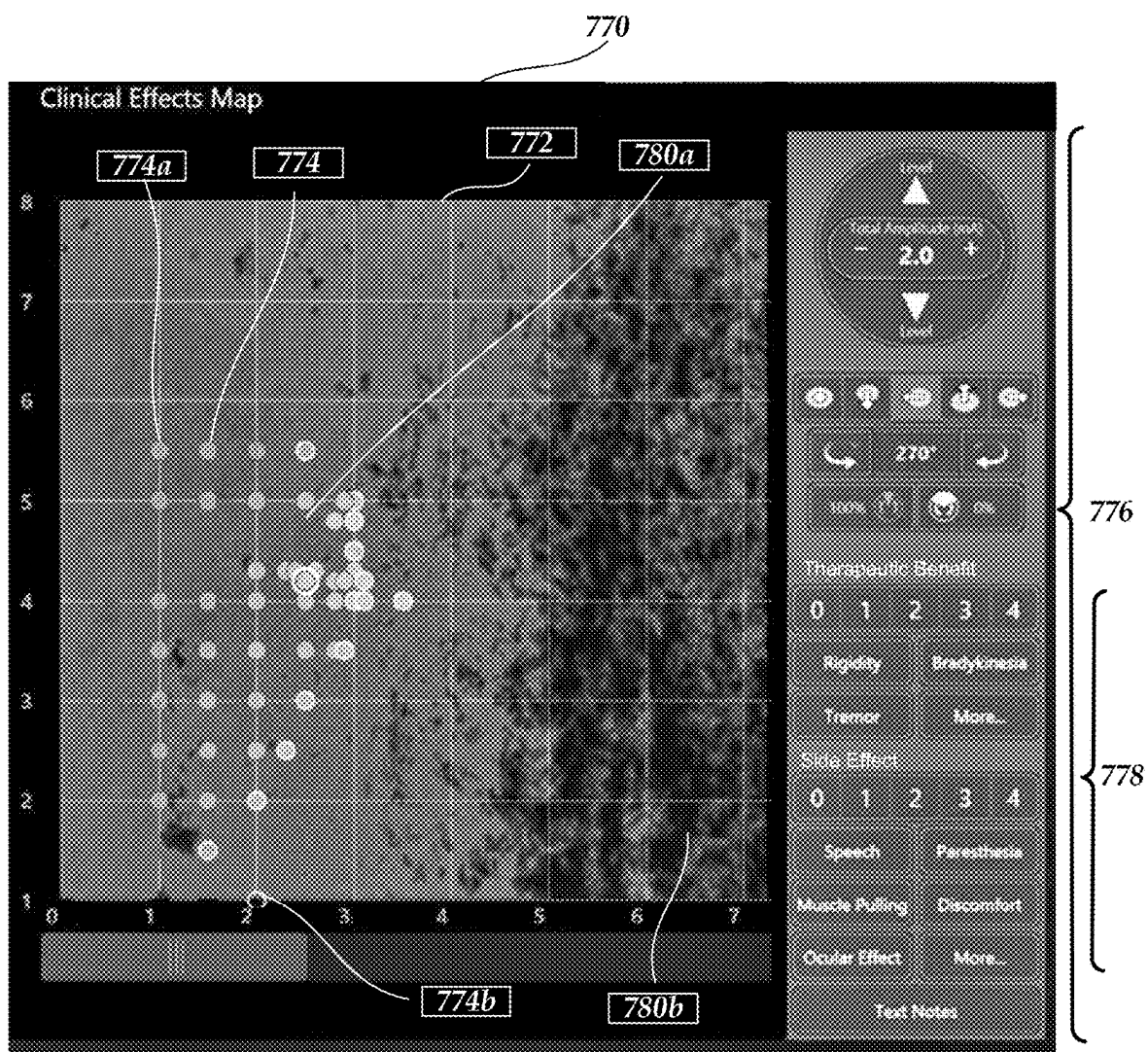


Fig. 7

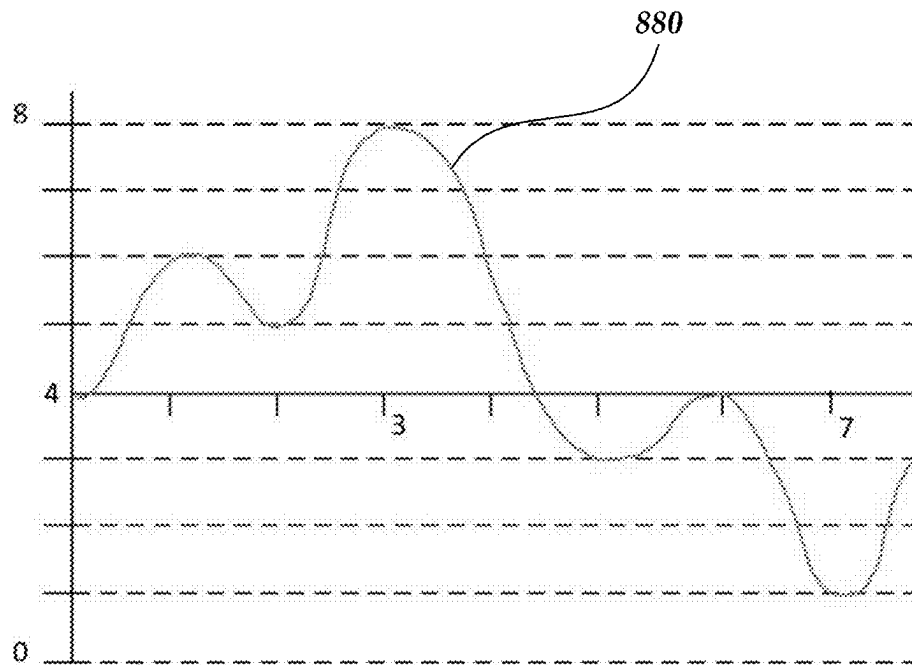


Fig. 8

1

SYSTEMS AND METHODS FOR AUTOMATED PROGRAMMING OF ELECTRICAL STIMULATION

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Patent Application Ser. No. 63/180,380, filed Apr. 27, 2021, which is incorporated herein by reference.

FIELD

The present disclosure is directed to the area of methods and systems for at least partially automating programming of electrical stimulation. The present disclosure is also directed to methods and systems for automating the testing of multiple sets of electrical stimulation parameters.

BACKGROUND

Implantable electrical stimulation systems have proven therapeutic in a variety of diseases and disorders. For example, deep brain stimulation systems have been used as a therapeutic modality for the treatment of Parkinson's disease, essential tremor, and the like.

Stimulators have been developed to provide therapy for a variety of treatments. A stimulator can include an implantable pulse generator (IPG), one or more leads, and an array of stimulator electrodes on each lead. The stimulator electrodes are in contact with or near the nerves, muscles, or other tissue to be stimulated. The pulse generator in the IPG generates electrical pulses that are delivered by the electrodes to body tissue.

Implantable medical devices (IMDs), including IPGs, typically have the capability to communicate data with an external device, such as a clinician programmer or a remote control, via a radio-frequency telemetry link or other wireless communication method. The clinician programmer can program the operating parameters of the implanted medical device. The remote control can switch programs. Modern implantable devices also include the capability for bidirectional communication so that information can be transmitted to the clinician programmer or remote control from the implanted device.

BRIEF SUMMARY

One aspect is a method for automating selection of stimulation parameters for a stimulation device implanted in a patient. The method includes setting, by a user, at least one limit on each of at least one stimulation parameter and performing, automatically using at least one processor, the following actions for each of a plurality of sets of the stimulation parameters constrained by the at least one limit: stimulating the patient, by the stimulation device, using the set of stimulation parameters, sensing one or more effects arising in response to the stimulation, and updating, by the at least one processor, a collection of the effects and sets of stimulation parameters with the one or more effects and the set of stimulation parameters. The method further includes selecting, by the processor, one of the sets of stimulation parameters based on the effects.

In at least some aspects, the method further includes, after selecting the one of the sets of stimulation parameters, fine-tuning, manually by the user, the stimulation parameters

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of the one of the sets of stimulation parameters. In at least some aspects, the method further includes programming the stimulation device with the one of the sets of stimulation parameters.

In at least some aspects, the sensing includes sensing a signal from the stimulated tissue. In at least some aspects, the signal is an evoked compound action potential (ECAP), evoked resonant neural activity (ERNA), or local field potential (LFP). In at least some aspects, at least one of the one or more effects is a feature of the signal, wherein the feature is a range of the signal, an area under a curve of the signal, a length of the curve of the signal, a maximum of the signal, or a minimum of the signal.

In at least some aspects, the performing includes performing the actions for sets of stimulation parameters at different axial locations along an electrical stimulation lead and selecting a one of the different axial locations based on the sensed one or more effects. In at least some aspects, the performing further includes, after selecting the one of the different axial locations, performing the actions for sets of stimulation parameters at second axial locations within one axial electrode spacing of the selected one of the different axial locations and selecting one of the second axial locations or the selected one of the different axial locations. In at least some aspects, the performing further includes, after selecting the one of the different axial locations, performing the actions for sets of stimulation parameters at different rotational directions around the electrical stimulation lead at the selected one of the different axial locations and selecting a one of the different rotational directions based on the sensed one or more effects. In at least some aspects, the performing further includes, after selecting the one of the different rotational directions, performing the actions for sets of stimulation parameters at second rotational directions within 120 degrees of the selected one of the different rotational directions and selecting one of the second rotational directions or the selected one of the different rotational directions. In at least some aspects, the performing further includes, after selecting the one of the different rotational directions, performing the actions for sets of stimulation parameters at different stimulation amplitudes for the selected one of the different rotational directions and selecting one of the stimulation amplitudes.

Another aspect is a system for selection of stimulation parameters for a stimulation device implanted in a patient. The system includes at least one processor configured to perform actions, the actions including receiving, from a user, at least one limit for each of at least one stimulation parameter and performing the following actions for each of a plurality of sets of the stimulation parameters constrained by the at least one limit: directing the stimulation device to stimulate the patient using the set of stimulation parameters, sensing one or more effects arising in response to the stimulation, and updating a collection of the effects and sets of stimulation parameters with the one or more effects and the set of stimulation parameters. The actions further include selecting one of the sets of stimulation parameters based on the effects.

In at least some aspects, the system further includes the stimulation device. In at least some aspects, the stimulation device includes an electrical stimulation lead having electrodes disposed along a distal portion of the electrical stimulation lead. In at least some aspects, sensing one or more effects including sensing the one or more effects arising in response to the stimulation using at least one of the electrodes of the electrical stimulation lead. In at least some aspects, the system further includes a sensor external to the

stimulation device, wherein sensing one or more effects including sensing the one or more effects arising in response to the stimulation using the sensor. In at least some aspects, the actions further include programming the stimulation device with the one of the sets of stimulation parameters.

Yet another aspect is a non-transitory computer-readable medium having stored thereon instructions for execution by at least one processor. The instructions include receiving, from a user, at least one limit for each of at least one stimulation parameter and performing the following actions for each of a plurality of sets of the stimulation parameters constrained by the at least one limit: directing the stimulation device to stimulate the patient using the set of stimulation parameters, sensing one or more effects arising in response to the stimulation, and updating a collection of the effects and sets of stimulation parameters with the one or more effects and the set of stimulation parameters. The instructions further including selecting one of the sets of stimulation parameters based on the effects.

In at least some aspects, the instructions further include, after selecting the one of the sets of stimulation parameters, fine-tuning, manually by the user, the stimulation parameters of the one of the sets of stimulation parameters. In at least some aspects, the instructions further include programming the stimulation device with the one of the sets of stimulation parameters. In at least some aspects, the instructions further include repeating the performing instruction periodically to track disease progression over time

BRIEF DESCRIPTION OF THE DRAWINGS

Non-limiting and non-exhaustive embodiments of the present invention are described with reference to the following drawings. In the drawings, like reference numerals refer to like parts throughout the various figures unless otherwise specified.

For a better understanding of the present invention, reference will be made to the following Detailed Description, which is to be read in association with the accompanying drawings, wherein:

FIG. 1 is a schematic view of one embodiment of an electrical stimulation system that includes one or more leads that can be coupled to an IPG;

FIG. 2 is a block diagram of elements of an electrical stimulation system;

FIG. 3A is a schematic perspective view of a distal portion of one embodiment of an electrical stimulation lead with segmented electrodes;

FIG. 3B is a schematic perspective view of a distal portion of another embodiment of an electrical stimulation lead with segmented electrodes;

FIG. 3C is a schematic perspective view of a distal portion of a third embodiment of an electrical stimulation lead with segmented electrodes;

FIG. 3D is a schematic perspective view of a distal portion of a fourth embodiment of an electrical stimulation lead with segmented electrodes;

FIG. 3E is a schematic perspective view of a distal portion of a fifth embodiment of an electrical stimulation lead with segmented electrodes;

FIG. 4 is a flowchart of one embodiment of a method of determining stimulation parameters for therapy;

FIG. 5 is a flowchart of one embodiment of a method for selection of a set of stimulation parameters;

FIG. 6 is a flowchart of one embodiment of a method of sensing a stimulation effect;

FIG. 7 is a schematic illustration of one embodiment of a user interface with a clinical effects map; and

FIG. 8 is schematic illustration of one embodiment of a signal measured using a sensor in response to stimulation.

DETAILED DESCRIPTION

The present disclosure is directed to the area of methods and systems for at least partially automating programming of electrical stimulation. The present disclosure is also directed to methods and systems for automating the testing of multiple sets of electrical stimulation parameters.

Implantable electrical stimulation systems and devices are used herein to exemplify the inventions, but it will be understood that these inventions can be utilized with other stimulation systems and devices. Examples of implantable electrical stimulation systems include, but are not limited to, a least one lead with one or more electrodes disposed along a distal end of the lead and one or more terminals disposed along the one or more proximal ends of the lead. Examples of electrical stimulation systems with leads are found in, for example, U.S. Pat. Nos. 6,181,969; 6,295,944; 6,391,985; 6,516,227; 6,609,029; 6,609,032; 6,741,892; 7,244,150; 7,450,997; 7,672,734; 7,761,165; 7,783,359; 7,792,590; 7,809,446; 7,949,395; 7,974,706; 8,831,742; 8,688,235; 8,175,710; 8,224,450; 8,271,094; 8,295,944; 8,364,278; and 8,391,985; U.S. Patent Application Publications Nos. 2007/0150036; 2009/0187222; 2009/0276021; 2010/0076535; 2010/0268298; 2011/0004267; 2011/0078900; 2011/0130817; 2011/0130818; 2011/0238129; 2011/0313500; 2012/0016378; 2012/0046710; 2012/0071949; 2012/0165911; 2012/0197375; 2012/0203316; 2012/0203320; 2012/0203321; 2012/0316615; 2013/0105071; 2011/0005069; 2010/0268298; 2011/0130817; 2011/0130818; 2011/0078900; 2011/0238129; 2011/0313500; 2012/0016378; 2012/0046710; 2012/0165911; 2012/0197375; 2012/0203316; 2012/0203320; and 2012/0203321, all of which are incorporated by reference in their entireties.

Turning to FIG. 1, one embodiment of an electrical stimulation system 10 includes one or more stimulation leads 12 and an implantable pulse generator (IPG) 14. The system 10 can also include one or more of an external remote control (RC) 16, a clinician's programmer (CP) 18, an external trial stimulator (ETS) 20, or an external charger 22. The IPG and ETS are examples of control modules for the electrical stimulation system.

The IPG 14 is physically connected, optionally via one or more lead extensions 24, to the stimulation lead(s) 12. Each lead carries multiple electrodes 26 arranged in an array. The IPG 14 includes pulse generation circuitry that delivers electrical stimulation energy in the form of, for example, a pulsed electrical waveform (i.e., a temporal series of electrical pulses) to the electrode array 26 in accordance with a set of stimulation parameters. The implantable pulse generator can be implanted into a patient's body, for example, below the patient's clavicle area or within the patient's abdominal cavity or at any other suitable site. The implantable pulse generator 14 can have multiple stimulation channels which may be independently programmable to control the magnitude of the current stimulus from each channel. In some embodiments, the implantable pulse generator 14 can have any suitable number of stimulation channels including, but not limited to, 4, 6, 8, 12, 16, 32, or more stimulation channels. The implantable pulse generator 14 can have one, two, three, four, or more connector ports, for receiving the terminals of the leads and/or lead extensions.

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The ETS 20 may also be physically connected, optionally via the percutaneous lead extensions 28 and external cable 30, to the stimulation leads 12. The ETS 20, which may have similar pulse generation circuitry as the IPG 14, also delivers electrical stimulation energy in the form of, for example, a pulsed electrical waveform to the electrode array 26 in accordance with a set of stimulation parameters. One difference between the ETS 20 and the IPG 14 is that the ETS 20 is often a non-implantable device that is used on a trial basis after the neurostimulation leads 12 have been implanted and prior to implantation of the IPG 14, to test the responsiveness of the stimulation that is to be provided. Any functions described herein with respect to the IPG 14 can likewise be performed with respect to the ETS 20.

The RC 16 may be used to telemetrically communicate with or control the IPG 14 or ETS 20 via a uni- or bi-directional wireless communications link 32. Once the IPG 14 and neurostimulation leads 12 are implanted, the RC 16 may be used to telemetrically communicate with or control the IPG 14 via a uni- or bi-directional communications link 34. Such communication or control allows the IPG 14, for example, to be turned on or off and to be programmed with different stimulation parameter sets. The IPG 14 may also be operated to modify the programmed stimulation parameters to actively control the characteristics of the electrical stimulation energy output by the IPG 14. In at least some embodiments, the CP 18 (or RC 16 or other programming device) allows a user, such as a clinician, the ability to program stimulation parameters for the IPG 14 and ETS 20 in the operating room and in follow-up sessions. Alternately, or additionally, in at least some embodiments, stimulation parameters can be programed via wireless communications (e.g., Bluetooth) between the RC 16 (or other external device such as a hand-held electronic device like a mobile phone, tablet, or the like) and the IPG 14.

The CP 18 may perform this function by indirectly communicating with the IPG 14 or ETS 20, through the RC 16, via a wireless communications link 36. Alternatively, the CP 18 may directly communicate with the IPG 14 or ETS 20 via a wireless communications link (not shown). In at least some embodiments, the stimulation parameters provided by the CP 18 are also used to program the RC 16, so that the stimulation parameters can be subsequently modified by operation of the RC 16 in a stand-alone mode (i.e., without the assistance of the CP 18). The CP 18 or RC 16 can be any suitable device including, but not limited to, a computer or other computing device, laptop, mobile device (for example, a mobile phone or tablet), or the like or any combination thereof. The CP 18 or RC 16 can include software applications for interacting with the IPG 14 or ETS 20 and for programming the IPG 14 or ETS 20.

Additional examples of the RC 16, CP 18, ETS 20, and external charger 22 can be found in the references cited herein as well as U.S. Pat. Nos. 6,895,280; 6,181,969; 6,516,227; 6,609,029; 6,609,032; 6,741,892; 7,949,395; 7,244,150; 7,672,734; and 7,761,165; 7,974,706; 8,175,710; 8,224,450; and 8,364,278; and U.S. Patent Application Publication No. 2007/0150036, all of which are incorporated herein by reference in their entireties.

FIG. 2 is a schematic overview of one embodiment of components of an electrical stimulation system 200 including an electronic subassembly 210 disposed within an IPG 14 (FIG. 1). It will be understood that the electrical stimulation system can include more, fewer, or different components and can have a variety of different configurations including those configurations disclosed in the stimulator references cited herein.

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The IPG 14 (FIG. 1) can include, for example, a power source 212, antenna 218, receiver 202, processor 204, and memory 205. Some of the components (for example, power source 212, antenna 218, receiver 202, processor 204, and memory 205) of the electrical stimulation system can be positioned on one or more circuit boards or similar carriers within a sealed housing of the IPG 14 (FIG. 1), if desired. Unless indicated otherwise, the term “processor” refers to both embodiments with a single processor and embodiments with multiple processors.

An external device, such as a CP or RC 206, can include a processor 207, memory 208, an antenna 217, and a user interface 219. The user interface 219 can include, but is not limited to, a display screen on which a digital user interface can be displayed and any suitable user input device, such as a keyboard, touchscreen, mouse, track ball, or the like or any combination thereof.

Any power source 212 can be used including, for example, a battery such as a primary battery or a rechargeable battery. Examples of other power sources include super capacitors, nuclear or atomic batteries, mechanical resonators, infrared collectors, thermally-powered energy sources, flexural powered energy sources, bioenergy power sources, fuel cells, bioelectric cells, osmotic pressure pumps, and the like including the power sources described in U.S. Pat. No. 7,437,193, incorporated herein by reference in its entirety.

As another alternative, power can be supplied by an external power source through inductive coupling via the antenna 218 or a secondary antenna. The external power source can be in a device that is mounted on the skin of the user or in a unit that is provided near the user on a permanent or periodic basis.

If the power source 212 is a rechargeable battery, the battery may be recharged using the antenna 218, if desired. Power can be provided to the battery for recharging by inductively coupling the battery through the antenna to a recharging unit 216 external to the user. Examples of such arrangements can be found in the references identified above.

In one embodiment, electrical current is emitted by the electrodes 26 on the lead body to stimulate nerve fibers, muscle fibers, or other body tissues near the electrical stimulation system. A processor 204 is generally included to control the timing and electrical characteristics of the electrical stimulation system. For example, the processor 204 can, if desired, control one or more of the timing, frequency, amplitude, width, and waveform of the pulses. In addition, the processor 204 can select which electrodes can be used to provide stimulation, if desired. In some embodiments, the processor 204 may select which electrode(s) are cathodes and which electrode(s) are anodes. In some embodiments, the processor 204 may be used to identify which electrodes provide the most useful stimulation of the desired tissue. Instructions for the processor 204 can be stored on the memory 205. Instructions for the processor 207 can be stored on the memory 208.

Any processor 204 can be used for the IPG and can be as simple as an electronic device that, for example, produces pulses at a regular interval or the processor can be capable of receiving and interpreting instructions from the CP/RC 206 (such as CP 18 or RC 16 of FIG. 1) that, for example, allows modification of pulse characteristics. In the illustrated embodiment, the processor 204 is coupled to a receiver 202 which, in turn, is coupled to the antenna 218. This allows the processor 204 to receive instructions from an external source to, for example, direct the pulse character-

istics and the selection of electrodes, if desired. Any suitable processor **207** can be used for the CP/RC **206**.

Any suitable memory **205**, **208** can be used including computer-readable storage media may include, but is not limited to, volatile, nonvolatile, non-transitory, removable, and non-removable media implemented in any method or technology for storage of information, such as computer readable instructions, data structures, program modules, or other data. Examples of computer-readable storage media include, but are not limited to, RAM, ROM, EEPROM, flash memory, or other memory technology/CD-ROM, digital versatile disks ("DVD") or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by a processor.

In one embodiment, the antenna **218** is capable of receiving signals (e.g., RF signals) from an antenna **217** of a CP/RC **206** (see, CP **18** or RC **16** of FIG. **1**) which is programmed or otherwise operated by a user. The signals sent to the processor **204** via the antenna **218** and receiver **202** can be used to modify or otherwise direct the operation of the electrical stimulation system. For example, the signals may be used to modify the pulses of the electrical stimulation system such as modifying one or more of pulse width, pulse frequency, pulse waveform, and pulse amplitude. The signals may also direct the electrical stimulation system **200** to cease operation, to start operation, to start signal acquisition, to stop signal acquisition, to start charging the battery, or to stop charging the battery. In other embodiments, the stimulation system does not include an antenna **218** or receiver **202** and the processor **204** operates as programmed.

Optionally, the electrical stimulation system **200** may include a transmitter (not shown) coupled to the processor **204** and the antenna **218** for transmitting signals back to the CP/RC **206** or another unit capable of receiving the signals. For example, the electrical stimulation system **200** may transmit signals indicating whether the electrical stimulation system **200** is operating properly or not or indicating when the battery needs to be charged or the level of charge remaining in the battery. The processor **204** may also be capable of transmitting information about the pulse characteristics so that a user or clinician can determine or verify the characteristics.

Transmission of signals can occur using any suitable method, technique, or platform including, but not limited to, inductive transmission, radiofrequency transmission, Bluetooth™, Wi-Fi, cellular transmission, near field transmission, infrared transmission, or the like or any combination thereof. In addition, the IPG **14** can be wirelessly coupled to the RC **16** or CP **18** using any suitable arrangement include direct transmission or transmission through a network, such as a local area network, wide area network, the Internet, or the like or any combination thereof. The CP **18** or RC **16** may also be capable of coupling to, and sending data or other information to, a network **220**, such as a local area network, wide area network, the Internet, or the like or any combination thereof.

At least some of the stimulation electrodes can take the form of segmented electrodes that extend only partially around the perimeter (for example, the circumference) of the lead. These segmented electrodes can be provided in sets of electrodes, with each set having electrodes circumferentially distributed about the lead at a particular longitudinal position.

In FIGS. **3A**, **3B**, and **3D** the electrodes are shown as including both ring electrodes **120** and segmented electrodes

122. In some embodiments, the electrodes are all segmented electrode **122**, as illustrated in FIGS. **3C** and **3E**. The segmented electrodes **122** of FIG. **3A** are in sets of three, where the three segmented electrodes of a particular set are electrically isolated from one another and are circumferentially offset along the lead **12**. Any suitable number of segmented electrodes can be formed into a set including, for example, two, three, four, or more segmented electrodes. The lead **12** of FIG. **3A** has thirty segmented electrodes **122** (ten sets of three electrodes each) and two ring electrodes **120** for a total of 32 electrodes.

Segmented electrodes can be used to direct stimulus current to one side, or even a portion of one side, of the lead. When segmented electrodes are used in conjunction with an implantable pulse generator that delivers current stimulus, current steering can be achieved to deliver the stimulus more precisely to a position around an axis of the lead (i.e., radial positioning around the axis of the lead). Segmented electrodes may provide for superior current steering than ring electrodes because target structures in deep brain stimulation are not typically symmetric about the axis of the distal electrode array. Instead, a target may be located on one side of a plane running through the axis of the lead. Through the use of a segmented electrode array, current steering can be performed not only along a length of the lead but also around a perimeter of the lead. This provides precise three-dimensional targeting and delivery of the current stimulus to neural target tissue, while potentially avoiding stimulation of other tissue.

FIG. **3A** illustrates a 32-electrode lead **12** with a lead body **106** and two ring electrodes **120** proximal to thirty segmented electrodes **122** arranged in ten sets of three segmented electrodes each. In the illustrated embodiments, the ring electrodes **120** are proximal to the segmented electrodes **122**. In other embodiments, the ring electrodes **120** can be proximal to, or distal to, or any combination thereof.

Any number of segmented electrodes **122** may be disposed on the lead body including, for example, one, two, three, four, five, six, seven, eight, nine, ten, eleven, twelve, thirteen, fourteen, fifteen, sixteen, twenty, twenty-four, twenty-eight, thirty, thirty-two, or more segmented electrodes **122**. It will be understood that any number of segmented electrodes **122** may be disposed along the length of the lead body. A segmented electrode **122** typically extends only 75%, 67%, 60%, 50%, 40%, 33%, 25%, 20%, 17%, 15%, or less around the circumference of the lead.

The segmented electrodes **122** may be grouped into sets of segmented electrodes, where each set is disposed around a circumference of the lead **12** at a particular longitudinal portion of the lead **12**. The lead **12** may have any number of segmented electrodes **122** in a given set of segmented electrodes. The lead **12** may have one, two, three, four, five, six, seven, eight, or more segmented electrodes **122** in a given set. The lead **12** may have any number of sets of segmented electrode including, but not limited to, one, two, three, four, five, six, eight, ten, twelve, fifteen, sixteen, twenty, or more sets. The segmented electrodes **122** may be uniform, or vary, in size and shape. In some embodiments, the segmented electrodes **122** are all of the same size, shape, diameter, width or area or any combination thereof. In some embodiments, the segmented electrodes **122** of each circumferential set (or even all segmented electrodes disposed on the lead **12**) may be identical in size and shape.

Each set of segmented electrodes **122** may be disposed around the circumference of the lead body to form a substantially cylindrical shape around the lead body. The spacing between individual electrodes of a given set of the

segmented electrodes may be the same, or different from, the spacing between individual electrodes of another set of segmented electrodes on the lead 12. In at least some embodiments, equal spaces, gaps or cutouts are disposed between each segmented electrode 122 around the circumference of the lead body. In other embodiments, the spaces, gaps or cutouts between the segmented electrodes 122 may differ in size or shape. In other embodiments, the spaces, gaps, or cutouts between segmented electrodes 122 may be uniform for a particular set of the segmented electrodes 122, or for all sets of the segmented electrodes 122. The sets of segmented electrodes 122 may be positioned in irregular or regular intervals along a length of the lead body.

The electrodes of the lead 12 are typically disposed in, or separated by, a non-conductive, biocompatible material of a lead body 106 including, for example, silicone, polyurethane, and the like or combinations thereof. The lead body 106 may be formed in the desired shape by any process including, for example, extruding, molding (including injection molding), casting, and the like. Electrodes and connecting wires can be disposed onto or within a lead body either prior to or subsequent to a molding or casting process. The non-conductive material typically extends from the distal end of the lead body 106 to the proximal end of the lead body 106.

FIG. 3B to 3E illustrate other embodiments of leads with segmented electrodes 122. FIG. 3B illustrates a sixteen electrode lead 12 having one ring electrode 120 that is proximal to five sets of three segmented electrodes 122 each. FIG. 3C illustrates a sixteen electrode lead 12 having eight sets of two segmented electrodes 122 each. As illustrated in FIG. 3C, an embodiment of a lead 12 does not necessarily include a ring electrode. FIG. 3D illustrates a sixteen electrode lead 12 having four ring electrodes 120 that are proximal to six sets of two segmented electrodes 122 each. FIG. 3E illustrates a thirty-two electrode lead 12 having sixteen sets of two segmented electrodes 122 each (for clarity of illustration, not all of the electrodes are shown). It will be recognized that any other electrode combination of ring electrodes, segmented electrodes, or both types of electrodes can be used.

When the lead 12 includes both ring electrodes 120 and segmented electrodes 122, the ring electrodes 120 and the segmented electrodes 122 may be arranged in any suitable configuration. For example, when the lead 12 includes two or more ring electrodes 120 and one or more sets of segmented electrodes 122, the ring electrodes 120 can flank the one or more sets of segmented electrodes 122. Alternately, the two or more ring electrodes 120 can be disposed proximal to the one or more sets of segmented electrodes 122 or the two or more ring electrodes 120 can be disposed distal to the one or more sets of segmented electrodes 122.

The electrodes 120, 122 may have any suitable longitudinal length including, but not limited to, 2, 3, 4, 4.5, 5, or 6 mm. The longitudinal spacing between adjacent electrodes 120, 122 may be any suitable amount including, but not limited to, 1, 2, or 3 mm, where the spacing is defined as the distance between the nearest edges of two adjacent electrodes. In some embodiments, the spacing is uniform between longitudinally adjacent electrodes along the length of the lead. In other embodiments, the spacing between longitudinally adjacent electrodes may be different or non-uniform along the length of the lead.

Examples of leads with segmented electrodes include U.S. Patent Application Publications Nos. 2010/0268298; 2011/0005069; 2011/0078900; 2011/0130803; 2011/0130816; 2011/0130817; 2011/0130818; 2011/0078900;

2011/0238129; 2011/0313500; 2012/0016378; 2012/0046710; 2012/0071949; 2012/0165911; 2012/0197375; 2012/0203316; 2012/0203320; 2012/0203321; 2013/0197602; 2013/0261684; 2013/0325091; 2013/0317587; 2014/0039587; 2014/0353001; 2014/0358209; 2014/0358210; 2015/0018915; 2015/0021817; 2015/0045864; 2015/0021817; 2015/0066120; 2013/0197424; 2015/0151113; 2014/0358207; and U.S. Pat. No. 8,483,237, all of which are incorporated herein by reference in their entireties. A lead may also include a tip electrode and examples of leads with tip electrodes include at least some of the previously cited references, as well as U.S. Patent Application Publications Nos. 2014/0296953 and 2014/0343647, all of which are incorporated herein by reference in their entireties. A lead with segmented electrodes may be a directional lead that can provide stimulation in a particular direction using the segmented electrodes.

Although electrical stimulation of tissue has demonstrated effectiveness for many medical conditions, there can be side-effects to electrical stimulation. Multi-electrode leads may enable greater selectivity of tissue to be stimulated which may provide improved therapy, but there can be a challenge in determining which electrodes and stimulation parameters provide therapeutic benefit while reducing or eliminating side effects.

In at least some instances, a treating physician may wish to tailor the stimulation parameters (such as which one or more of the stimulating electrodes to use, the stimulation pulse amplitude (such as current or voltage amplitude depending on the stimulator being used,) the stimulation pulse width, the stimulation frequency, the duty cycle, the stimulation phase, or the like or any combination thereof) for a particular patient. Conventionally, the clinician or programmer tries different selections of electrodes, stimulation amplitude, and other parameters and then determines therapy effectiveness or side effects using visual observation, sensor observation, patient feedback, or the like. This can be a lengthy and time-consuming process.

In contrast to the conventional programming methods, as described herein, an automated (or at least partially automated) programming system or method can include the automated sequential testing of different sets of stimulation parameters and evaluation of each set of stimulation parameters using a sensor or sensed responses to the stimulation. The automated programming system or method can select one or more of the tested sets of stimulation parameters for therapy. In at least some embodiments, the automation can then be fine-tuned (or otherwise altered) by manual adjustment of the stimulation parameter by a clinician or other programmer.

FIG. 4 illustrates one embodiment of a method of determining stimulation parameters for therapy. In step 402, a user, such as a clinician or other programmer, sets limits on one or more stimulation parameters. Examples of stimulation parameters that can be limited include, but are not limited to, amplitude, electrode selection, stimulation direction, pulse width, pulse frequency, or the like or any combination thereof. For example, the user may place a lower limit on the amplitude because lower amplitudes are unlikely to produce satisfactory effects or the user may place an upper limit because higher amplitudes are likely to produce side effects. The user may place limits on which electrodes are selected on the lead (for example, electrodes 2 to 5 in an eight-electrode array) so that only electrodes that are near the target region are selected. For leads with segmented electrodes, the user may limit the direction of the stimulation so that only certain segmented electrodes (within

those limits on direction) will be tested. This may reflect knowledge of the position and orientation of the lead in the tissue and the desired target region for stimulation.

Steps **404** to **412** form a loop in which a processor automatically tests multiple sets of stimulation parameters, constrained by the limits set in step **402**, and determines the effect(s) of the test stimulations. The effect(s) can be therapeutic effect(s) or side effect(s) or any combination thereof. In at least some embodiments, the loop is closed and occurs automatically. In at least some embodiments, the loop may occur with no manual input. In at least some embodiments, the loop may occur with manual input limited to input from the clinician, programmer, or patient regarding observed therapeutic or side effects. In at least some embodiments, the loop can be halted manually by the user, clinician, programmer, patient, or any combination thereof. This manual control can be used, for example, to halt the process if there are safety or patient comfort concerns or issues. In at least some embodiments, the processor may also include one or more thresholds (for example, threshold values of side effects) that, when met or exceeded, may halt the loop as well. In at least some embodiments, halting the loop based on one or more thresholds can result from the sensing of therapeutic or side effects described below.

In step **404**, a set of stimulation parameters is selected. In at least some embodiments, one or more (or even all of the) sets of stimulation parameters that are to be tested are determined prior to the start of the initial loop of steps **404** to **412**. In at least some embodiments, additional sets of stimulation parameters may be determined during, or as a result of, the loop of steps **404** to **412** based, for example, on the effect(s) sensed during the testing. Non-limiting examples of procedures for determining sets of parameters based on feedback or sensed effects are described in U.S. Pat. No. 10,603,498, which is incorporated herein by reference in its entirety.

In at least some embodiments, the testing of the sets of stimulation parameters can be performed according to a predefined routine or order. For example, the predefined routine or order for testing different electrode selections may start at a proximal position and move distally for each subsequent set or start at a distal position and move proximally for each subsequent set. A predefined routine or order for testing different amplitudes may start at a low or intermediate amplitude and increase for each subsequent set.

In other embodiments, the testing of the sets of stimulation parameters can be performed in any other suitable order including random, pseudo-random, partially random, or based on any other selection method or criterion/criteria.

In at least some embodiments, the testing of the sets of stimulation parameters can be performed using coarse testing of sets that differ in one parameter, followed by fine-tuning for that parameter, and then proceeding to do the same for one or more additional parameters. Examples of methods for testing the sets of stimulation parameters are presented below in the embodiments illustrated in FIG. 5.

In step **406**, the patient is stimulated using the selected set of stimulation parameters. In step **408**, one or more effects of the stimulation are sensed. Any suitable therapeutic effects, side effects, response effects, or any combination thereof can be sensed. A response effect can be any effect that is a result of the stimulation and can include, but are not limited to, therapeutic (e.g., beneficial) effects or side (e.g., detrimental) effects. Such response effects can be, for example, neural response such as an evoked compound action potential (ECAP), an evoked resonant neural activity

(ERNA), or a local field potential (LFP). These response effects are not necessarily therapeutic or side effects.

In at least some embodiments, one or more effects are sensed by a sensor external to the IPG, by a sensor coupled to the IPG, or by one or more of the electrodes of the leads (acting as sensor(s)), or the like or any combination thereof. In at least some embodiments, the sensor or electrodes are used to obtain measurements of the effect(s). In at least some embodiments, the IPG, CP, RC, ETS, or other device are used to evaluate the measurements from the sensor or electrodes. Examples of methods for sensing the effect(s) are presented below including the embodiments illustrated in FIG. 6, as well as in U.S. Pat. Nos. 10,357,657; 10,369,364; and 10,716,505 and U.S. Patent Application Publications Nos. 2014/0243926; 2014/0276707; 2014/0277282; 2020/0376263; 2020/0398057; and 2021/0023374, all of which are incorporated herein by reference in their entireties.

In at least some embodiments, the valuation of the measurements from the sensor or electrodes can include a comparative evaluation of the therapeutic effect(s) and side effect(s). In at least some embodiments, an intensity or benefits scale can be used to rate the therapeutic effect(s) and an intensity or detriment scale can be used to rate the side effect(s). In at least some embodiments, these two scales can be selected for comparison. For example, in at least some embodiments, a therapeutic score may be determined by subtracting the rating for the side effect(s) from the rating for the therapeutic effect(s). The higher the therapeutic score, the likelihood that the corresponding stimulation is beneficial. Other algorithms or methods of comparing the therapeutic effect(s) to the side effect(s) can be used.

In step **410**, a collection of the effects and sets of stimulation parameters are updated with the current set of stimulation parameters and the sensed effect(s). The collection can be in any suitable form including, but not limited to, a database, a clinical effects map or graph, or the like or any combination thereof. FIG. 7 illustrates one embodiment of a clinical effects map. Additional information regarding clinical effects maps can be found in, for example, U.S. Pat. Nos. 9,227,074; 9,248,296; 9,358,398; 9,474,903; 10,603,498; and 10,071,249 and U.S. Patent Application Publication No. 2018/0264278, all of which are incorporated herein in their entireties.

FIG. 7 illustrates one embodiment of a user interface **770** that includes a clinical effects map **772** that maps the electrode position along the y-axis and stimulation amplitude (in mA) along the x-axis. Examples of three-dimensional clinical effects maps and clinical effects maps using cylindrical coordinates (for example, where rho represent amplitude, phi represents a rotational angle, and z represents electrode position) are presented in the references cited above. Positions on the clinical effects map **772** correspond to stimulations, each associated with activation of an actual or virtual electrode at a particular stimulation amplitude. The virtual electrodes can correspond to stimulation using a combination of actual electrodes. For example, a position between electrode 1 and electrode 2 can correspond to stimulation using a combination of electrodes 1 and 2 (for example, 50% of the stimulation amplitude on electrode 1 and 50% of the stimulation amplitude on electrode 2 to provide a virtual electrode with a midpoint between electrodes 1 and 2).

In the illustrated embodiment, the markings **774** on the clinical effects map **772** correspond to actual stimulation events. In at least some embodiments, the markings **774** may also provide information about the effect(s) produced by the stimulation. In the illustrated embodiment of FIG. 7, the

markings can have a circle **774a** that corresponds to therapeutic effect(s) with variation in color or intensity indicating the strength of the therapeutic effect(s). The markings can have a ring **774b** that corresponds to side effect(s) with variation in color or intensity indicating the strength of the side effect(s). In the illustrated embodiment, the user interface **770** includes an input section **776** with controls **778** for inputting specific types of therapeutic or side effects and corresponding intensities.

The clinical effects map **772** may also include a feature that indicates potential target regions **780a** and potential avoidance regions **780b** that are determined based on the preceding stimulation instances. For example, the potential target regions **780a** can be indicated on the clinical effects map **772** as a particular background color (for example, green) and correspond to regions near stimulation instances in which the therapeutic effect(s) are greater than the side effect(s) (and, optionally, where the side effect(s) are below a threshold level). The potential avoidance regions **780b** can be indicated on the clinical effects map **772** as a different background color (for example, red) and correspond to regions where the trendlines from the preceding stimulation instances indicate that the side effect(s) are likely greater than the therapeutic effect(s) or where the side effects are likely to meet or exceed a threshold level. The potential avoidance region **780b** may also include the region out the limits set by the user in step **402**, as described above.

In step **412**, the processor determines if there is another set of stimulation parameters to test. If yes, then the method returns to step **404**. If no, then the method continues.

In step **414**, the processor selects one of the sets of stimulation parameters based on the testing. In at least some embodiments, the set of stimulation parameters with the best therapeutic response is selected. In at least some embodiments, the processor will select the set of stimulation parameters based on both the therapeutic response and the presence or severity of side effects. In at least some embodiments, the ratings of the therapeutic effect(s), side effect(s), or any combination thereof can be used to select the set of stimulation parameters. In at least some embodiments, the therapeutic scores (e.g., the different between the rating for the therapeutic effect(s) and the rating for the side effect(s)) for the sets of stimulation parameters can be used to select the set of stimulation parameters.

In optional step **416**, a clinician or programmer can fine-tune the stimulation parameters by altering or modifying the set of stimulation parameters selected by the processor in step **414**. In at least some embodiments, the processor may require that the clinician or programmer authorize or accept the selected set of stimulation parameters. In step **418**, the IPG is programmed using the selected set of stimulation parameters or the set of stimulation parameters after modification or alteration by the clinician or programmer in step **416**.

In at least some embodiments, steps **404** to **412** can be performed while the patient is asleep when the testing is performed without patient or clinician feedback. In at least some embodiments, the method can result in personalized therapy for the patient or can provide programming that is tailored to the patient's physiological response.

In at least some embodiments, the method can be performed periodically and can be used to track disease progression over time. In at least some embodiments, the method can be performed periodically to adjust stimulation parameters or to improve closed-loop device operation.

One embodiment of a method for selection of a set of stimulation parameters is illustrated in FIG. 5. In step **502**,

a number (for example, two, three, four, five, six, eight, ten, twelve, or more) of different axial positions along the lead are tested. Each of the axial positions can correspond to a single electrode (or a set of segmented electrodes at the same axial position along the lead) or to multiple electrodes (including multiple electrodes at different axial positions along the lead to provide a virtual electrode that is a combination of these electrodes.) As an example, an axial position between electrode 1 (or a set of segmented electrodes at axial position 1 along the lead) and electrode 2 (or a set of segmented electrodes at axial position 2 along the lead) can correspond to stimulation using a combination of electrodes 1 and 2. For example, an axial position of 1.5 may correspond to 50% of the stimulation amplitude on electrode 1 and 50% of the stimulation amplitude on electrode 2 and an axial position of 1.25 may correspond to 75% of the stimulation amplitude on electrode 1 and 25% of the stimulation amplitude on electrode 2.

In at least some embodiments, the different axial positions are selected to be uniformly spread along the portion of the lead that is permitted for use. In at least some embodiments, when a particular axial position produces side effects over a threshold level or produces therapeutic effects below a particular threshold, other axial positions near, distal, or proximal to the particular axial position may be precluded from testing. The processor may determine that the precluded axial positions may produce side effects or not produce substantial therapeutic effects.

The processor selects one or more of these initial axial positions for proceeding. For example, the processor may use a comparative evaluation of the therapeutic effect(s) and side effect(s). In at least some embodiments, an intensity or benefits scale can be used to rate the therapeutic effect(s) and an intensity or detriment scale can be used to rate the side effect(s). In at least some embodiments, these two scales can be selected for comparison. For example, in at least some embodiments, a therapeutic score may be determined by subtracting the rating for the side effect(s) from the rating for the therapeutic effect(s). The higher the therapeutic score, the likelihood that the corresponding stimulation is beneficial. Other algorithms or methods of comparing the therapeutic effect(s) to the side effect(s) can be used.

In optional step **504**, the processor may fine-tune the axial position by testing additional axial positions near the selected initial axial positions. For example, if the axial position corresponding to electrode 3 (or a set of segmented electrodes at axial position 3 along the lead) is selected, then fine-tuning may include testing sets of stimulation parameters with axial positions in the range of 2 to 4 (e.g., 2.25, 2.5, 2.75, 3.25, 3.5, and 3.75) which corresponds axial positions within one axial electrode spacing of axial position 3. In at least some embodiments, the fine-tuning includes testing axial positions within one, two, three, or four or more axial electrode spacings from the selected initial axial position. The processor then selects one or more these fine-tuned axial positions (or the initial axial positions) for proceeding using any suitable method including those described above.

In optional step **506**, when the selected initial or fine-tuned axial positions include segmented electrodes, the processor can select two or more different rotational directions at the selected axial positions. Each of the rotational directions can correspond to a single segmented electrode or to multiple segmented electrodes (including multiple segmented electrodes at different axial positions along the lead to provide a virtual electrode that is a combination of these segmented electrodes.) As an example, a rotational direction between segmented electrode 1 and segmented electrode 2

can correspond to stimulation using a combination of segmented electrodes 1 and 2. For example; a rotational direction of 1.5 may correspond to 50% of the stimulation amplitude on segmented electrode 1 and 50% of the stimulation amplitude on segmented electrode 2.

The processor selects one or more of these initial rotational directions for proceeding. In optional step 508, the processor can also fine-tune the rotational direction(s) by testing additional rotational directions near the selected initial rotational direction. For example, if the rotational direction corresponding to electrode 2 is selected, then fine-tuning may include testing sets of stimulation parameters with rotational directions in the range of 1 to 3 (e.g., 1.25, 1.5, 1.75, 2.25, 2.5, and 2.75). In at least some embodiments, the fine-tuning includes testing rotational directions within 120, 100, 90, 60, 45, or fewer degrees from the selected initial rotational direction. One of the fine-tuned rotational directions (or initial rotational direction(s)) can be selected for proceeding using any suitable method including those described above.

In step 510, one or more additional stimulation parameters can be tested using the selected axial position and optional rotational direction. For example, different stimulation amplitudes can be tested. Other stimulation parameters can include pulse duration, pulse frequency, or the like or any combination thereof. For each of these one or more additional stimulation parameters, a value can be selected based on the testing using any suitable method including those described above.

One embodiment of a method of sensing a stimulation effect is provided in FIG. 6. In step 602, after stimulating with a set of stimulation parameters, the sensor makes a measurement of one or more effect(s) of the stimulation including, but not limited to, therapeutic effects, side effects, response effects, or the like or any combination thereof. The stimulation effect can be, for example, an evoked compound action potential (ECAP), an evoked resonant neural activity (ERNA), a local field potential (LFP), or the like or any combination thereof. In some embodiments, the stimulation effect can be an effect of the disease or disorder such as rigidity, tremor, impaired or unwanted movement, synchronized signals in the brain or elsewhere, seizures, or the like or any combination thereof. The sensor can be, for example, a sensor external to the IPG, by a sensor coupled to the IPG, or by one or more of the electrodes of the leads (acting as sensor(s)), or the like or any combination thereof. In at least some embodiments, there may be at least some digital signal processing of the measurement in step 602, such as, for example, finite impulse response (FIR) filtering or recursively averaging epochs of data.

In step 604, one or more features of the measurements are extracted. For example, in at least some embodiments, the measurements can be signal that is recorded over time and one or more features of the signal can be extracted. FIG. 8 illustrates one embodiment of a signal 880 measured using a sensor (such as an external or internal sensor or one or more of the electrodes of the lead.) In FIG. 8, the signal 880 is presented as a graph of current (or voltage) amplitude (y-axis) versus time (x-axis). Examples of the measurement of such signals 880 can be found at, for example, U.S. Pat. Nos. 10,357,657; 10,369,364; and 10,716,505 and U.S. Patent Application Publications Nos. 2014/0243926; 2014/0276707; 2014/0277282; 2020/0376263; 2020/0398057; and 2021/0023374, all of which are incorporated herein by reference in their entireties.

Non-limiting examples of features that can be extracted from the signal 880 include, for example, the range of

amplitude, value(s) of positive peaks, value(s) of negative peaks, area under the signal curve, length of the signal curve, or the like or any combination thereof. In at least some embodiments, a range of time is selected or predefined for the extraction of the feature(s). For example, the range of time may be a block of 64, 128, 256, or 512 (or any other suitable number) of samples of the measurements made by the sensor.

Signal 880 of FIG. 8 can be used as an example, to illustrate different measurements. For example, a y-range measurement can correspond to $y_{max}-y_{min}$ which in the case of signal 880 is $8-1=7$. An x-range measurement can correspond, for example, to the x-range between y_{max} (e.g., $x_{y,max}=3$ for signal 880) and x_{min} (e.g., $x_{y,min}=7$ for signal 880) which in the case of signal 880 is $(7-3)=4$.

Another example is the area under the curve of the signal. This area can be determined using any suitable method including integration techniques or the like. One example uses the algorithm Area Under Curve= $\sum|y|$ at each value of x which for signal 880 is $6+5+8+6+3+4+1=33$.

Yet another example, is the length of the curve of the signal. This length can be determined using any suitable method for determining or estimating a length of a curve. One example uses the algorithm Curve Length= $\sum|\Delta y|=\sum|y_i-y_{i-1}|$ which for signal 880 is $2+1+3+2+3+1+3=15$.

The methods, techniques, and systems described herein are presented in the context of an electrical stimulation system, but it will be recognized that these methods, techniques, and systems can be used with an optical stimulation system or an electrical/optical stimulation system. Examples of optical stimulation systems or electrical/optical stimulation systems are found in U.S. Patent Application Publications Nos. 2013/0317572; 2013/0317573; 2017/0259078; 2017/0225007; 2018/0110971;

2018/0369606; 2018/0369607; 2019/0209849; 2019/0209834; 2020/0094047; and 2020/0155854 and U.S. patent application Ser. No. 16/883,404, all of which are incorporated herein by reference in their entireties.

It will be understood that each block of the flowchart illustration, and combinations of blocks in the flowchart illustration and methods disclosed herein, can be implemented by computer program instructions. These program instructions may be provided to a processor to produce a machine or engine, such that the instructions, which execute on the processor, create means for implementing the actions specified in the flowchart block or blocks or engine disclosed herein. The computer program instructions may be executed by a processor to cause a series of operational steps to be performed by the processor to produce a computer implemented process. The computer program instructions may also cause at least some of the operational steps to be performed in parallel. Moreover, some of the steps may also be performed across more than one processor, such as might arise in a multi-processor computing device. In addition, one or more processes may also be performed concurrently with other processes, or even in a different sequence than illustrated without departing from the scope or spirit of the invention.

The computer program instructions can be stored on any suitable computer-readable medium including, but not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks ("DVD") or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by a com-

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puting device. The computer program instructions can be stored locally or nonlocally (for example, in the Cloud).

The above specification and examples provide a description of the arrangement and use of the invention. Since many embodiments of the invention can be made without departing from the spirit and scope of the invention, the invention also resides in the claims hereinafter appended.

What is claimed as new and desired to be protected is:

1. A method for automating selection of stimulation parameters for a stimulation device implanted in a patient, the method comprising:

setting, by a user, at least one limit on each of at least three stimulation parameters, each of the at least three stimulation parameters being different and selected from an axial position along an electrical stimulation lead, a rotational direction of stimulation, a stimulation amplitude, a pulse duration, a pulse frequency, or a duty cycle, wherein each of the at least one limit comprises an upper limit, a lower limit, a limit on electrode selection, or a limit on stimulation direction; and

performing, automatically using at least one processor, the following actions sequentially and individually for each of the at least three stimulation parameters constrained by the at least one limit, wherein the actions are fully performed with respect to one of the at least three stimulation parameters before performing the actions for another one of the at least three stimulation parameters:

selecting a one of the at least three stimulation parameters,

stimulating tissue of the patient, by the stimulation device, using different first values for the selected one of the at least three stimulation parameters constrained by the at least one limit for the one of the at least three stimulation parameters,

for each of the different first values, sensing one or more first effects arising in response to the corresponding stimulation including sensing a signal from the stimulated tissue, wherein the signal is an evoked compound action potential (ECAP), evoked resonant neural activity (ERNA), or local field potential (LFP), wherein at least one of the one or more first effects is a feature of the signal over a selected range of time, wherein the feature comprises at least one of a range of amplitude of the signal over the selected range of time, an amount of time between a maximum amplitude of the signal and a minimum amplitude of the signal for the selected range of time, an area under a curve of the signal for the selected range of time, or a length of the curve of the signal for the selected range of time,

updating, by the at least one processor and for each of the different first values, a collection of effects and corresponding values of the at least three stimulation parameters with the one or more first effects and the corresponding different first values of the one of the at least three stimulation parameters,

selecting, by the processor, a coarse value from the different first values of the one of the at least three stimulation parameters based on the sensed one or more first effects,

stimulating tissue of the patient, by the stimulation device, using different second values for the selected one of the at least three stimulation parameters that are within a predefined relative range around the selected coarse value of the one of the at least three

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stimulation parameters constrained by the at least one limit for the one of the at least three stimulation parameters,

for each of the different second values, sensing one or more second effects arising in response to the stimulation including sensing a signal from the stimulated tissue, wherein the signal is an evoked compound action potential (ECAP), evoked resonant neural activity (ERNA), or local field potential (LFP), wherein at least one of the one or more second effects is a feature of the signal over a selected range of time, wherein the feature comprises at least one of a range of amplitude of the signal over the selected range of time, an amount of time between a maximum amplitude of the signal and a minimum amplitude of the signal for the selected range of time, an area under a curve of the signal for the selected range of time, or a length of the curve of the signal for the selected range of time,

updating, by the at least one processor and for each of the different second values, a collection of effects and corresponding values of the at least three stimulation parameters with the one or more second effects and the corresponding different second values of the one of the at least three stimulation parameters, and selecting, by the processor, a fine-tuned value from the different second values of the one of the at least three stimulation parameters based on the sensed one or more second effects,

wherein, after the fine-tuned value is selected for any one of the at least three stimulation parameters, that fine-tuned value is used in subsequent stimulations when performing the actions for another one of the at least three stimulation parameters.

2. The method of claim 1, further comprising, after selecting the fine-tuned value of the one of the at least three stimulation parameters, further fine-tuning, manually by the user, to obtain a manual fine-tuned value for one of the at least three stimulation parameters.

3. The method of claim 1, further comprising programming the stimulation device with the selected fine-tuned values for each of the at least three stimulation parameters.

4. The method of claim 1, wherein the signal is an evoked compound action potential (ECAP) or an evoked resonant neural activity (ERNA).

5. The method of claim 1, wherein the at least three stimulation parameters comprises a first stimulation parameter corresponding to an axial location along the electrical stimulation lead.

6. The method of claim 5, wherein the predefined relative range is one axial electrode spacing on both sides of the selected coarse value of the axial location.

7. The method of claim 5, wherein the at least three stimulation parameters comprises a second stimulation parameter corresponding to a rotational direction around the electrical stimulation lead.

8. The method of claim 7, wherein the predefined relative range is 120 degrees on both sides of the selected coarse value of the rotational direction.

9. The method of claim 7, wherein the at least three stimulation parameters comprises a third stimulation parameter corresponding to a stimulation amplitude.

10. A system for selection of stimulation parameters for a stimulation device implanted in a patient, the system comprising:

at least one processor configured to perform actions, the actions comprising:

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receiving, from a user, at least one limit for each of at least three stimulation parameters, each of the at least three stimulation parameters being different and selected from an axial position along an electrical stimulation lead, a rotational direction of stimulation, a stimulation amplitude, a pulse duration, a pulse frequency, or a duty cycle, wherein each of the at least one limit comprises an upper limit, a lower limit, a limit on electrode selection, or a limit on stimulation direction; and

performing the following actions sequentially and individually for each of the at least three stimulation parameters constrained by the at least one limit, wherein the actions are fully performed with respect to one of the at least three stimulation parameters before performing the actions for another one of the at least three stimulation parameters:

selecting a one of the at least three stimulation parameters,

directing the stimulation device to stimulate tissue of the patient using different first values for the selected one of the at least three stimulation parameters constrained by the at least one limit for the one of the at least three stimulation parameters,

for each of the different first values, sensing one or more first effects arising in response to the corresponding stimulation including sensing a signal from the stimulated tissue, wherein the signal is an evoked compound action potential (ECAP), evoked resonant neural activity (ERNA), or local field potential (LFP), wherein at least one of the one or more effects is a feature of the signal over a selected range of time, wherein the feature comprises at least one of a range of amplitude of the signal over the selected range of time, an amount of time between a maximum amplitude of the signal and a minimum amplitude of the signal for the selected range of time, an area under a curve of the signal for the selected range of time, or a length of the curve of the signal for the selected range of time,

updating, for each of the different first values, a collection of the effects and corresponding values of the at least three stimulation parameters with the one or more first effects and the corresponding different first values of the one of the at least three stimulation parameters,

selecting a coarse value from the different first values of the one of the at least three stimulation parameters based on the sensed one or more first effects,

directing stimulation of tissue of the patient, by the stimulation device, using different second values for the selected one of the at least three stimulation parameters that are within a predefined relative range around the selected coarse value of the one of the at least three stimulation parameters constrained by the at least one limit for the one of the at least three stimulation parameters,

for each of the different second values, sensing one or more second effects arising in response to the stimulation including sensing a signal from the stimulated tissue, wherein the signal is an evoked compound action potential (ECAP), evoked resonant neural activity (ERNA), or local field potential (LFP), wherein at least one of the one or more second effects is a feature of the signal over a selected range of

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time, wherein the feature comprises at least one of a range of amplitude of the signal over the selected range of time, an amount of time between a maximum amplitude of the signal and a minimum amplitude of the signal for the selected range of time, an area under a curve of the signal for the selected range of time, or a length of the curve of the signal for the selected range of time,

updating, by the at least one processor and for each of the different second values, a collection of effects and corresponding values of the at least three stimulation parameters with the one or more second effects and the corresponding different second values of the one of the at least three stimulation parameters, and selecting, by the processor, a fine-tuned value from the different second values of the one of the at least three stimulation parameters based on the sensed one or more second effects,

wherein, after the fine-tuned value is selected for any one of the at least three stimulation parameters, that fine-tuned value is used in subsequent stimulations when performing the actions for another one of the at least three stimulation parameters.

11. The system of claim 10, further comprising the stimulation device.

12. The system of claim 11, wherein the stimulation device comprises the electrical stimulation lead having electrodes disposed along a distal portion of the electrical stimulation lead.

13. The system of claim 12, wherein sensing one or more first effects comprising sensing the one or more first effects arising in response to the stimulation using at least one of the electrodes of the electrical stimulation lead.

14. The system of claim 10, further comprising a sensor external to the stimulation device, wherein sensing one or more first effects comprising sensing the one or more first effects arising in response to the stimulation using the sensor.

15. The system of claim 10, wherein the actions further comprise programming the stimulation device with the fine-tuned values selected for the at least three stimulation parameters.

16. A non-transitory computer-readable medium having stored thereon instructions for execution by at least one processor, the instructions comprising:

a) receiving, from a user, at least one limit for each of at least three stimulation parameters, each of the at least three stimulation parameters being different and selected from an axial position along an electrical stimulation lead, a rotational direction of stimulation, a stimulation amplitude, a pulse duration, a pulse frequency, or a duty cycle, wherein each of the at least one limit comprises an upper limit, a lower limit, a limit on electrode selection, or a limit on stimulation direction; and

b) performing the following actions sequentially and individually for each of the at least three stimulation parameters constrained by the at least one limit, wherein the actions are fully performed with respect to one of the at least three stimulation parameters before performing the actions for another one of the at least three stimulation parameters:

selecting a one of the at least three stimulation parameters,

directing the stimulation device to stimulate tissue of the patient using different first values for the selected one of the at least three stimulation parameters

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constrained by the at least one limit for the one of the at least three stimulation parameters,

for each of the different first values, sensing one or more first effects arising in response to the corresponding stimulation including sensing a signal from the stimulated tissue, wherein the signal is an evoked compound action potential (ECAP), evoked resonant neural activity (ERNA), or local field potential (LFP), wherein at least one of the one or more effects is a feature of the signal over a selected range of time, wherein the feature comprises at least one of a range of amplitude of the signal over the selected range of time, an amount of time between a maximum amplitude of the signal and a minimum amplitude of the signal for the selected range of time, an area under a curve of the signal for the selected range of time, or a length of the curve of the signal for the selected range of time,

updating, for each of the different first values, a collection of the effects and corresponding values of the at least three stimulation parameters with the one or more first effects and the corresponding different first values of the one of the at least three stimulation parameters,

selecting a coarse value from the different first values of the one of the at least three stimulation parameters based on the sensed one or more first effects,

directing stimulation of tissue of the patient, by the stimulation device, using different second values for the selected one of the at least three stimulation parameters that are within a predefined relative range around the selected coarse value of the one of the at least three stimulation parameters constrained by the at least one limit for the one of the at least three stimulation parameters,

for each of the different second values, sensing one or more second effects arising in response to the stimulation including sensing a signal from the stimulated tissue, wherein the signal is an evoked compound

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action potential (ECAP), evoked resonant neural activity (ERNA), or local field potential (LFP), wherein at least one of the one or more second effects is a feature of the signal over a selected range of time, wherein the feature comprises at least one of a range of amplitude of the signal over the selected range of time, an amount of time between a maximum amplitude of the signal and a minimum amplitude of the signal for the selected range of time, an area under a curve of the signal for the selected range of time, or a length of the curve of the signal for the selected range of time,

updating, by the at least one processor and for each of the different second values, a collection of effects and corresponding values of the at least three stimulation parameters with the one or more second effects and the corresponding different second values of the one of the at least three stimulation parameters, and selecting, by the processor, a fine-tuned value from the different second values of the one of the at least three stimulation parameters based on the sensed one or more second effects,

wherein, after the fine-tuned value is selected for any one of the at least three stimulation parameters, that fine-tuned value is used in subsequent stimulations when performing the actions for another one of the at least three stimulation parameters.

17. The non-transitory computer-readable medium of claim **16**, wherein the instructions further comprise, after selecting the fine-tuned value of the one of the at least three stimulation parameters, further fine-tuning, manually by the user, to obtain a manual fine-tuned value for one of the at least three stimulation parameters.

18. The non-transitory computer-readable medium of claim **16**, wherein the instructions further comprising repeating the instruction b) periodically to track disease progression over time.

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